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Subject: Data Analytics with R

Experiment No. 03

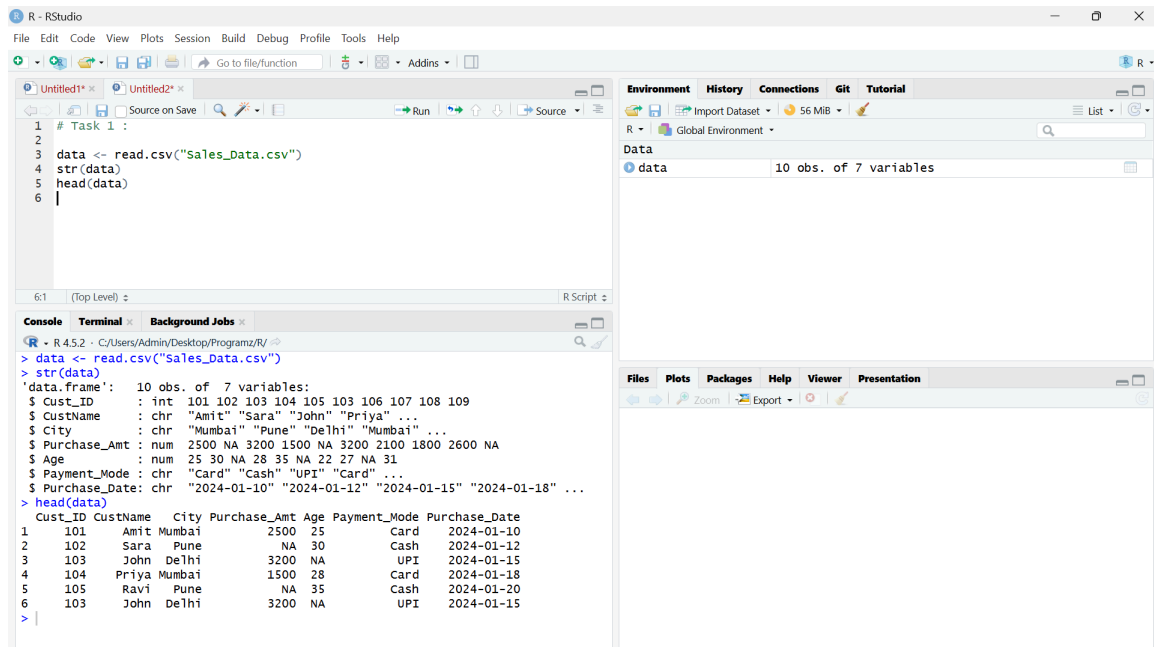
To implement a program to develop user-defined functions in R to perform data transformation operations

Title: To implement a program for user-defined functions in R for perform data transformation operations using built-in packages for mathematical operations

Aim:

To create and implement user-defined functions in R for performing various data transformation operations and mathematical computations using built-in R packages

Code / Outputs:



The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for reading a CSV file and displaying its structure and first rows.
- Environment:** Shows the 'data' object with 10 observations and 7 variables.
- Console:** Displays the execution output of the R code.

```
# Task 1 :  
1  
2  
3 data <- read.csv("Sales_Data.csv")  
4 str(data)  
5 head(data)  
6
```

```
> data <- read.csv("Sales_Data.csv")  
> str(data)  
'data.frame': 10 obs. of 7 variables:  
 $ Cust_ID : int 101 102 103 104 105 103 106 107 108 109  
 $ CustName : chr "Amit" "Sara" "John" "Priya" ...  
 $ City : chr "Mumbai" "Pune" "Delhi" "Mumbai" ...  
 $ Purchase_Amt : num 2500 NA 3200 1500 NA 3200 2100 1800 2600 NA  
 $ Age : num 25 30 NA 28 35 NA 22 27 NA 31  
 $ Payment_Mode : chr "Card" "Cash" "UPI" "Card" ...  
 $ Purchase_Date : chr "2024-01-10" "2024-01-12" "2024-01-15" "2024-01-18" ...  
> head(data)  
  Cust_ID CustName City Purchase_Amt Age Payment_Mode Purchase_Date  
1 101 Amit Mumbai 2500 25 Card 2024-01-10  
2 102 Sara Pune NA 30 Cash 2024-01-12  
3 103 John Delhi 3200 NA UPI 2024-01-15  
4 104 Priya Mumbai 1500 28 Card 2024-01-18  
5 105 Ravi Pune NA 35 Cash 2024-01-20  
6 103 John Delhi 3200 NA UPI 2024-01-15
```

R - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Source on Save Run Source

1 # Task 1 :
2
3 data <- read.csv("Sales_Data.csv")
4 str(data)
5 head(data)
6
7 # Task 2:
8
9 is.na(data)
10 sum(is.na(data))

10:17 (Top Level) R Script

Console Terminal Background Jobs

R - R 4.5.2 - C:/Users/Admin/Desktop/Programz/R/

```
2 102 Sara Pune NA 30 Cash 2024-01-12  
3 103 John Delhi 3200 NA UPI 2024-01-15  
4 104 Priya Mumbai 1500 28 Card 2024-01-18  
5 105 Ravi Pune NA 35 Cash 2024-01-20  
6 103 John Delhi 3200 NA UPI 2024-01-15  
7  
8 > is.na(data)  
9      Cust_ID CustName City Purchase_Amt Age Payment_Mode Purchase_Date  
10 [1,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
11 [2,] FALSE FALSE FALSE TRUE FALSE FALSE FALSE  
12 [3,] FALSE FALSE FALSE FALSE TRUE FALSE FALSE  
13 [4,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
14 [5,] FALSE FALSE FALSE TRUE FALSE FALSE FALSE  
15 [6,] FALSE FALSE FALSE FALSE TRUE FALSE FALSE  
16 [7,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
17 [8,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
18 [9,] FALSE FALSE FALSE FALSE TRUE FALSE FALSE  
19 [10,] FALSE FALSE FALSE TRUE FALSE FALSE FALSE  
20  
21 > sum(is.na(data))  
22 [1] 6  
23  
24 > |
```

Environment History Connections Git Tutorial

R - Global Environment

Data

data 10 obs. of 7 variables

Files Plots Packages Help Viewer Presentation

Zoom Export

R - RStudio

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Go to file/function Addins

Source on Save Run Source

7 # Task 2:
8
9 is.na(data)
10 sum(is.na(data))
11
12 # Task 3:
13 # handling missing data
14
15 data\$Age[is.na(data\$Age)] <- mean(data\$Age, na.rm = TRUE)
16 data\$Purchase_Amt[is.na(data\$Purchase_Amt)] <- median(data\$Purchase_Amt, na.rm = TRUE)
17

17:1 (Top Level) R Script

Console Terminal Background Jobs

R - R 4.5.2 - C:/Users/Admin/Desktop/Programz/R/

```
> data$Age[is.na(data$Age)] <- mean(data$Age, na.rm = TRUE)  
> data$Purchase_Amt[is.na(data$Purchase_Amt)] <- median(data$Purchase_Amt, na.rm = TRUE)  
> |
```

Environment History Connections Git Tutorial

R - Global Environment

Data

data 10 obs. of 7 variables

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Zoom Export

R - RStudio

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Go to file/function Addins

Source on Save Run Source

```
13 # handling missing data
14
15 data$Age[is.na(data$Age)] <- mean(data$Age, na.rm = TRUE)
16 data$Purchase_Amt[is.na(data$Purchase_Amt)] <- median(data$Purchase_Amt, n
17
18
19 # Task 4 :
20 # Remove duplicate records
21 dim(data)
22 data <- unique(data)
23 dim(data)
24
```

21:1 (Top Level) R Script

Console Terminal Background Jobs

R - R 4.5.2 - C:/Users/Admin/Desktop/Programz/R/

```
> dim(data)
[1] 9 7
> data <- unique(data)
> dim(data)
[1] 9 7
> |
```

Environment History Connections Git Tutorial

Import Dataset 114 MiB

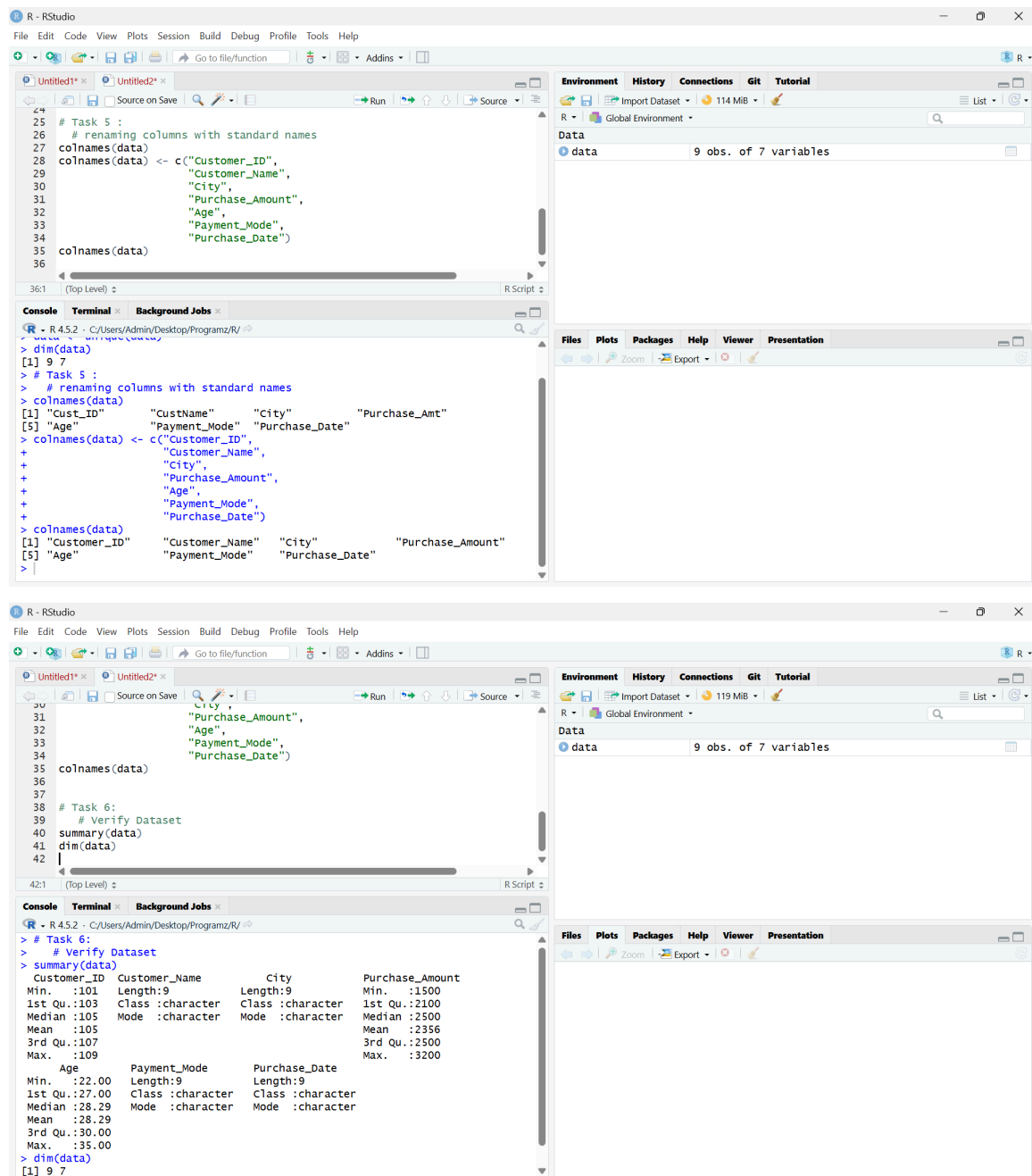
R - Global Environment

Data

data 9 obs. of 7 variables

Files Plots Packages Help Viewer Presentation

Zoom Export



Conclusion:

The experiment demonstrated how to import and clean datasets using R. students learned to identify missing values, remove duplicates and rename columns effectively. Data cleaning ensures improved data quality and reliability, forming a crucial preprocessing step before performing analytical or predictive tasks